Sampling Methods & Bias

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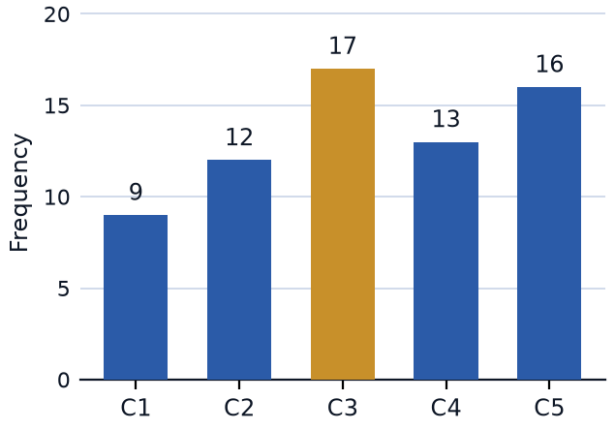
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Identifying sampling methods

A school assigns each student a number from 1 to 800, then uses a random number generator to select 40 numbers, surveying those students. Identify this sampling method.

2 [Foundational] • Target: Identifying sampling methods

A researcher divides a school into grade levels (9th, 10th, 11th, 12th), then takes a simple random sample of 20 students from each grade. Identify this sampling method and explain why it might be preferred over an SRS of the whole school.



3 [Foundational] • Target: Cluster sampling and systematic sampling

A researcher randomly selects 5 entire homeroom classes out of 40 in a school, and surveys every student in those 5 classes. Identify this sampling method, and contrast it with stratified sampling.

4 [Foundational] • Target: Cluster sampling and systematic sampling

A quality-control inspector checks every 25th item coming off a production line. Identify this sampling method.

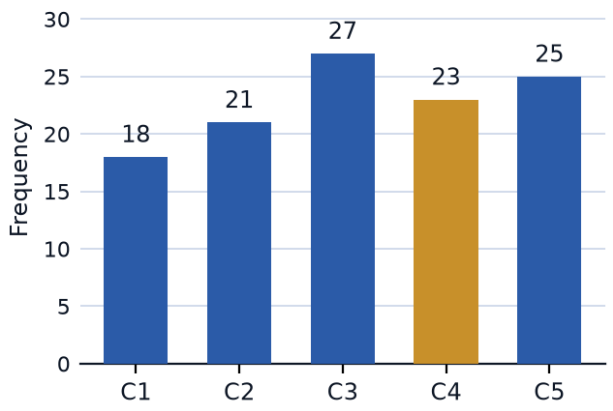
5 [Foundational] • Target: Sources of bias

A survey about a new school policy is voluntarily filled out online, and only 12% of students respond. Identify the type of bias most likely present, and explain why the results may not represent the whole student body.

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6 [Foundational] • Target: Sources of bias

A phone survey about internet usage habits is conducted using only landline phone numbers. Identify the type of bias this introduces.



7 [Foundational] • Target: Wording and response bias

A survey question reads: "Don't you agree that the wasteful new cafeteria policy should be reversed?" Explain why this question is problematic, and rewrite it in a neutral form.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Wording and response bias

Explain why respondents might not answer a sensitive question (such as about illegal activity) truthfully in a face-to-face interview, and name this type of bias.

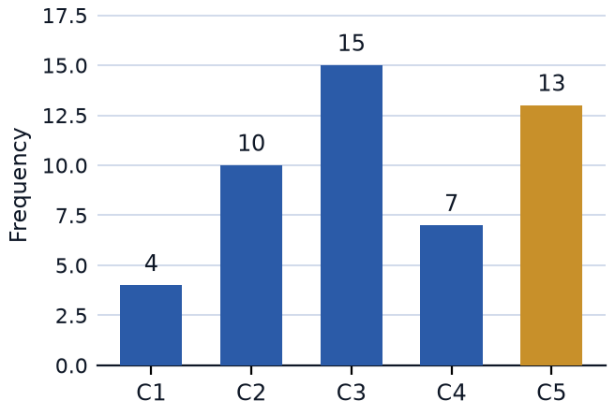
9 [Intermediate] • Target: Comparing sample size and bias

Explain why increasing the sample size does not fix a biased sampling method, using the voluntary response survey from earlier as an example.

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10 **[Intermediate]** • **Target: Designing a sampling plan**

A city has 4 distinct neighborhoods with very different income levels, and the mayor wants to survey residents about a new tax proposal in a way that fairly represents each neighborhood. Recommend a sampling method, and describe how it would be carried out.



11 **[Intermediate]** • **Target: Convenience sampling and its pitfalls**

A researcher stands outside a gym and surveys the first 50 people who walk out about their exercise habits, intending to represent the whole city's exercise habits. Identify this sampling method and explain the flaw in generalizing its results to the whole city.

12 **[Intermediate]** • **Target: Comparing sampling methods side by side**

A national retailer wants to estimate average customer satisfaction across all 500 of its stores. Compare how a cluster sample (randomly selecting 20 whole stores and surveying all customers there) and a stratified sample (grouping stores by region, then sampling customers from every region) would each be carried out, and state one advantage of each approach.

13 **[Intermediate]** • **Target: Identifying the population and sample**

A university wants to know the average commute time for all 30,000 enrolled students, and surveys a random sample of 400 students. Identify the population and the sample in this scenario.

14 **[Intermediate]** • **Target: Identifying the population and sample**

A national news outlet runs a text-in poll during a live broadcast, asking viewers to text "YES" or "NO" on a controversial policy question, and reports the results as representing "public opinion." Identify every source of bias you can find in this scenario, and explain why the reported percentages should not be trusted as an accurate reflection of the general public's views.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

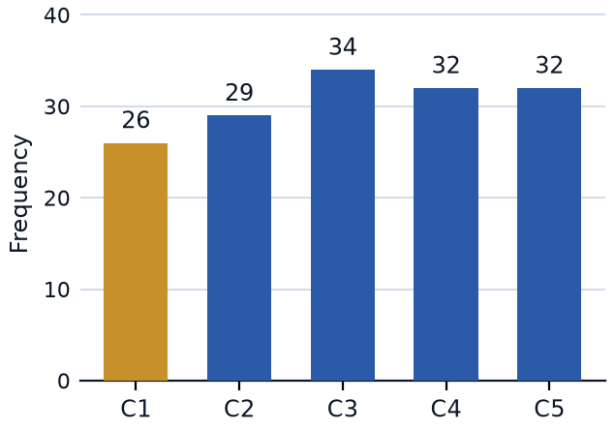
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15 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A hospital wants to estimate the average patient satisfaction score across its 12 different departments (e.g. cardiology, pediatrics, emergency), which are expected to have very different typical satisfaction levels. Describe, step by step, how you would design a stratified random sample to estimate overall patient satisfaction, and explain why a simple random sample of all patients might be less informative for comparing satisfaction across departments.

16 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A researcher wants to estimate the average time it takes customers to be served at a chain of 200 coffee shops nationwide. Due to budget constraints, the researcher can only visit a small number of physical locations. Describe how a cluster sample could be carried out in this situation, and explain a key risk of cluster sampling that would not apply if the researcher instead used a stratified sample based on region.



17 [Challenging] • Target: Comparing sampling methods side by side • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A school district with 12 high schools wants to sample 300 teachers. Compare: (a) Stratified sampling (taking an SRS of 25 teachers from each of the 12 schools) vs. (b) Cluster sampling (randomly selecting 2 schools and surveying all 150 teachers in each). Discuss sampling variability and logistical practicality.

18 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 1.01 Measures of Center & Spread

Level 3 · AdvancedA university alumni association randomly selects 500 graduates from its directory and sends them a survey asking for their current annual salary. Only 95 completed surveys are returned, yielding an average reported salary of 115,000. (a) Name the primary source of bias present. (b) Explain the likely direction of the bias on the estimated mean salary. (c) Explain whether the results can be generalized to all graduates of the university.

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19 [Challenging] • Target: Identifying sampling methods • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

Level 1 · FoundationsA random sample of 100 high school seniors finds that 64 plan to attend a four-year college. Construct a 95% confidence interval for the true population proportion p of seniors attending a four-year college.

20 [Challenging] • Target: Cluster sampling and systematic sampling • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A sample of size 100 from a population (known $\sigma = 20$) gives $\bar{x} = 142$. Find a 99% CI for μ and state what "99% confidence" means.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Identifying sampling methods

A factory assigns each worker a number from 1 to 1223, then uses a random number generator to select 70 numbers, surveying those workers. Identify this sampling method.

a. Convenience sample

b. Simple random sample (SRS)

c. Stratified random sample

d. Cluster sample

22 [Foundational] • Target: Cluster sampling and systematic sampling

A researcher divides a factory into groups (morning shift, afternoon shift, and night shift), then takes a simple random sample of 53 workers from each group. Identify this sampling method.

a. Cluster sample

b. Convenience sample

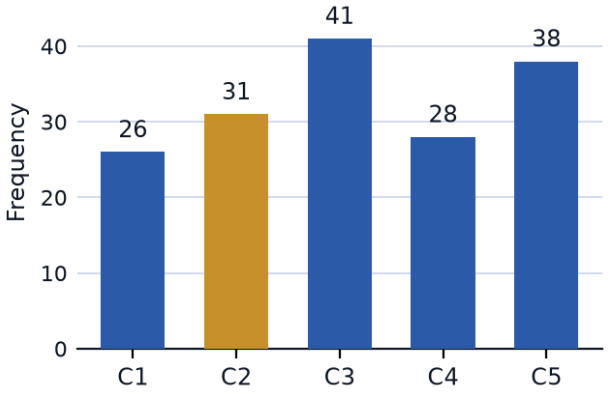
c. Stratified random sample

d. Systematic sample

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23 [Foundational] • Target: Sources of bias

A researcher randomly selects 3 entire delivery routes out of 33 at a shipping company, and surveys every member within those selected delivery routes. Identify this sampling method.



a. Stratified random sample

b. Simple random sample (SRS)

c. Systematic sample

d. Cluster sample

24 [Foundational] • Target: Wording and response bias

A quality-control inspector at a factory checks every 49th item coming off a production line, starting from a randomly chosen point among the first 49 items. Identify this sampling method.

a. Simple random sample (SRS)

b. Convenience sample

c. Systematic sample

d. Stratified random sample

25 [Foundational] • Target: Comparing sample size and bias

A researcher stands outside a farmers market and surveys the first 44 people encountered, intending to represent the general population. Identify this sampling method.

a. Systematic sample

b. Cluster sample

c. Simple random sample (SRS)

d. Convenience sample

26 [Foundational] • Target: Designing a sampling plan

A company assigns each employee a number from 1 to 1017, then uses a random number generator to select 61 numbers, surveying those employees. Identify this sampling method.

a. Simple random sample (SRS)

b. Convenience sample

c. Cluster sample

d. Systematic sample

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27 [Foundational] • Target: Convenience sampling and its pitfalls

A researcher divides a hospital into groups (department A, department B, and department C), then takes a simple random sample of 72 patients from each group. Identify this sampling method.

a. Cluster sample

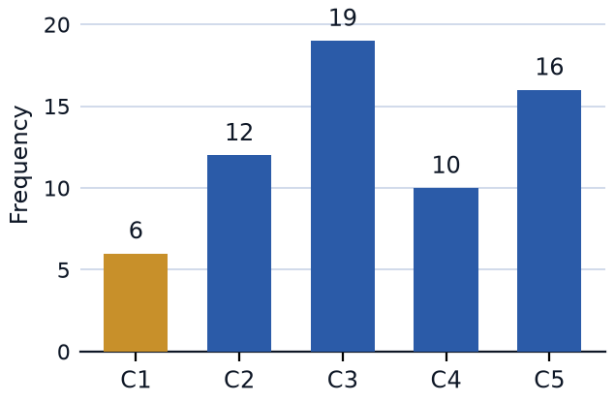
b. Stratified random sample

c. Convenience sample

d. Simple random sample (SRS)

28 [Foundational] • Target: Comparing sampling methods side by side

A researcher randomly selects 6 entire wards out of 18 at a hospital, and surveys every member within those selected wards. Identify this sampling method.



a. Stratified random sample

b. Convenience sample

c. Systematic sample

d. Cluster sample

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Identifying the population and sample

A researcher stands outside a train station and surveys the first 49 people encountered, intending to represent the general population. Identify this sampling method.

a. Cluster sample

b. Simple random sample (SRS)

c. Convenience sample

d. Systematic sample

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30 [Intermediate] • Target: Identifying sampling methods

A hospital assigns each patient a number from 1 to 1075, then uses a random number generator to select 80 numbers, surveying those patients. Identify this sampling method.

a. Systematic sample

b. Simple random sample (SRS)

c. Stratified random sample

d. Cluster sample

31 [Intermediate] • Target: Cluster sampling and systematic sampling

A researcher divides a factory into groups (freshmen, sophomores, juniors, and seniors), then takes a simple random sample of 67 workers from each group. Identify this sampling method.

a. Cluster sample

b. Systematic sample

c. Stratified random sample

d. Simple random sample (SRS)

32 [Intermediate] • Target: Sources of bias

A researcher randomly selects 9 entire homeroom classes out of 15 at a school, and surveys every member within those selected homeroom classes. Identify this sampling method.

a. Simple random sample (SRS)

b. Systematic sample

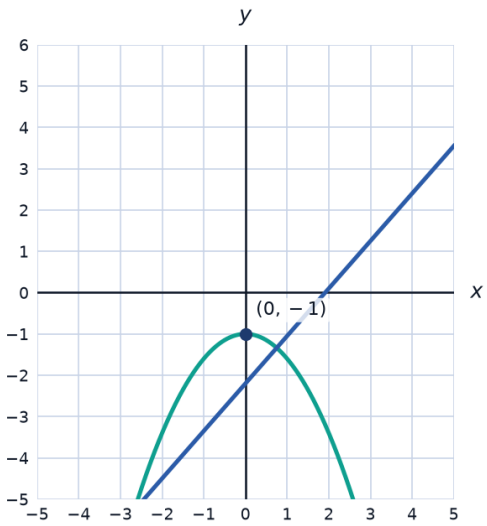
c. Cluster sample

d. Convenience sample

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33 [Intermediate] • Target: Wording and response bias

A quality-control inspector at a clinic checks every 32nd item coming off a production line, starting from a randomly chosen point among the first 32 items. Identify this sampling method.



a. Simple random sample (SRS)

b. Convenience sample

c. Systematic sample

d. Cluster sample

34 [Intermediate] • Target: Comparing sample size and bias

A survey about a new dress code is voluntarily filled out online by residents, and only 20% respond. Identify the type of bias most likely present.

a. Wording bias (leading question)

b. Convenience sampling bias

c. Voluntary response bias

d. Response bias (social desirability)

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35 **[Intermediate]** • **Target: Designing a sampling plan**

A phone survey about university services is conducted using only landline phone numbers, systematically excluding residents without internet access. Identify the type of bias this introduces.

a. Voluntary response bias

b. Response bias (social desirability)

c. Undercoverage bias

d. Wording bias (leading question)

36 **[Intermediate]** • **Target: Convenience sampling and its pitfalls**

Respondents are asked directly, face-to-face, about underage drinking. Identify the type of bias likely to affect their answers.

a. Wording bias (leading question)

b. Voluntary response bias

c. Convenience sampling bias

d. Response bias (social desirability)

37 **[Intermediate]** • **Target: Comparing sampling methods side by side**

A survey question reads: "Don't you agree that the wasteful dress code should be changed?" Identify the type of bias in this question.

a. Convenience sampling bias

b. Voluntary response bias

c. Wording bias (leading question)

d. Response bias (social desirability)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Identifying the population and sample • Cross-Lesson: 1.01 Measures of Center & Spread

A researcher surveys people at a farmers market and generalizes the results to the entire city's population. Identify the type of bias most likely present.

a. Response bias (social desirability)

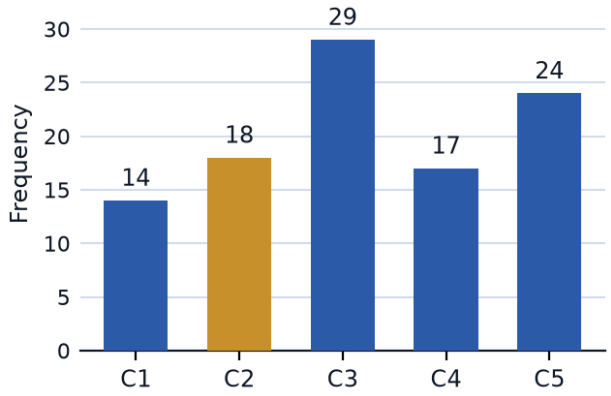
b. Convenience sampling bias

c. Voluntary response bias

d. Undercoverage bias

39 [Challenging] • Target: Identifying sampling methods • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A survey about a new policy is voluntarily filled out online by employees, and only 14% respond. Identify the type of bias most likely present.



a. Convenience sampling bias

b. Undercoverage bias

c. Voluntary response bias

d. Response bias (social desirability)

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40 [Challenging] • Target: Cluster sampling and systematic sampling • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A phone survey about university services is conducted using only landline phone numbers, systematically excluding people who don't check email. Identify the type of bias this introduces.

a. Response bias (social desirability)

b. Convenience sampling bias

c. Voluntary response bias

d. Undercoverage bias

41 [Challenging] • Target: Sources of bias • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Respondents are asked directly, face-to-face, about tax evasion. Identify the type of bias likely to affect their answers.

a. Undercoverage bias

b. Response bias (social desirability)

c. Wording bias (leading question)

d. Convenience sampling bias

42 [Challenging] • Target: Wording and response bias • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A survey question reads: "Don't you agree that the dangerous parking policy should be changed?" Identify the type of bias in this question.

a. Undercoverage bias

b. Voluntary response bias

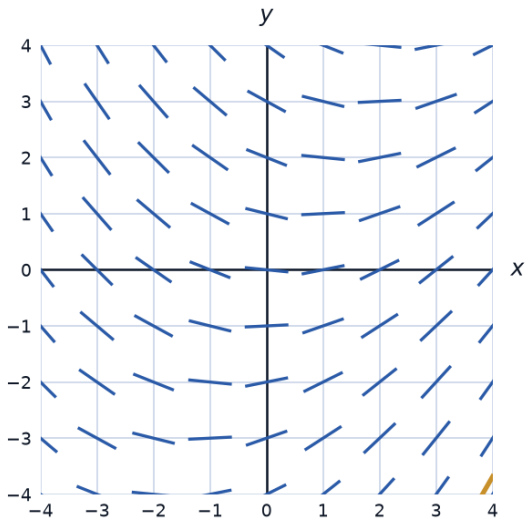
c. Wording bias (leading question)

d. Response bias (social desirability)

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43 [Challenging] • Target: Comparing sample size and bias • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A researcher surveys people at a shopping mall entrance and generalizes the results to the entire city's population. Identify the type of bias most likely present.



a. Undercoverage bias

b. Response bias (social desirability)

c. Wording bias (leading question)

d. Convenience sampling bias

44 [Challenging] • Target: Designing a sampling plan • Cross-Lesson: 3.03 The Normal Distribution

A survey about a proposed schedule change is voluntarily filled out online by employees, and only 8% respond. Identify the type of bias most likely present.

a. Voluntary response bias

b. Response bias (social desirability)

c. Wording bias (leading question)

d. Convenience sampling bias

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45 [Challenging] • Target: Convenience sampling and its pitfalls • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A phone survey about school services is conducted using only landline phone numbers, systematically excluding cell-phone-only households. Identify the type of bias this introduces.

a. Convenience sampling bias

b. Wording bias (leading question)

c. Undercoverage bias

d. Voluntary response bias

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